ThreatMatrix AI — Global Context Document v1.0

> **PURPOSE:** Paste this document into ANY AI model/chat session to instantly carry full project context. This is the single source of truth for cross-chat integration. Any task outlined in the Master Documentation can be started from this document.

> **LAST UPDATED:** 2026-02-25 | **PHASE:** Implementation (Week 1 / Day 1)

STRICT RULES FOR ANY MODEL RECEIVING THIS DOCUMENT

1. You are helping build **ThreatMatrix AI** — an AI-powered network anomaly detection and cyber threat intelligence system.
2. This is a **senior project** at HiLCoE School Of Computer Science & Technology, BUT it must be **enterprise-grade and sellable**.
3. **DO NOT** deviate from the architecture, stack, or scope defined below. If a task contradicts this document, flag it.
4. **DO NOT** suggest Kafka, Kubernetes, Elasticsearch, or any overengineered infrastructure. We use Docker Compose on a single VPS.
5. **DO NOT** suggest paid mapping APIs (Mapbox, Google Maps). We use Deck.gl + Maplibre GL (free, open-source).
6. **DO NOT** add features not listed in the 10 modules below. Scope is locked.
7. All code must be **production-quality** — typed, error-handled, documented. No prototyping shortcuts.
8. Follow the file structure defined in Section 4 exactly. Do not reorganize.
9. When writing Python, use **type hints, Pydantic models, async/await**. When writing TypeScript, use **strict mode**.
10. The UI must follow the **War Room / Intelligence Agency** design language. Dark theme only. Glassmorphism. Cyan accents.

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1. PROJECT IDENTITY

- **Name:** ThreatMatrix AI
- **Tagline:** "Intelligent Defense. Autonomous Vigilance."
- **Domain:** `threatmatrix-ai.vercel.app` (frontend) / VPS IP (backend API)
- **Team:** 4 members (Lead Architect, Full-Stack Dev, Business Manager, Tester)
- **Timeline:** Feb 24 → Apr 20, 2026 (8 weeks)
- **Current Phase:** Week 1 — Foundation

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2. TECHNOLOGY STACK (LOCKED — DO NOT CHANGE)

Layer	Technology	Version
Frontend	Next.js (App Router)	16.x
Language (FE)	TypeScript	5.x (strict)

Layer	Technology	Version
Styling	Vanilla CSS (CSS Variables)	—
Charts	Recharts	Latest
Maps	Deck.gl + Maplibre GL JS	Latest
Animations	Framer Motion	Latest
i18n	next-intl	Latest
Backend	FastAPI (Python)	0.115+
Language (BE)	Python	3.11+
Database	PostgreSQL	16
Cache/PubSub	Redis	7
ORM	SQLAlchemy	2.x
Migrations	Alembic	Latest
Packet Capture	Scapy	Latest
ML (Classical)	scikit-learn	Latest
ML (Deep)	TensorFlow/Keras	2.x
PDF Reports	ReportLab	Latest
Containerization	Docker + Docker Compose	Latest
Reverse Proxy	Nginx	Alpine
SSL	Let's Encrypt (certbot)	—
Frontend Hosting	Vercel (free tier)	—
Backend Hosting	VPS (Ubuntu 22.04, owned)	—

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3. ARCHITECTURE (THREE-TIER)

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┌──────────────────────────────────────────────────────────────────┐
│ TIER 1: CAPTURE ENGINE (Python/Scapy)                            │
│ • Sniffs packets on VPS network interface                        │
│ • Aggregates into flows (5-tuple: src_ip, dst_ip,                │
│   src_port, dst_port, protocol)                                  │
│ • Extracts 40+ features per flow                                 │
│ • Publishes to Redis channel: flows:live                         │
│ • Persists to PostgreSQL: network_flows table                    │
├──────────────────────────────────────────────────────────────────┤
│ TIER 2: INTELLIGENCE ENGINE (FastAPI)                            │
│ • REST API (40+ endpoints) + WebSocket server                    │
│ • ML Worker: subscribes to flows:live, runs 3 models,            │
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│   computes ensemble score, creates alerts                    │
│ • LLM Gateway: routes to DeepSeek/GLM/Groq with caching    │
│ • Threat Intel: syncs OTX, AbuseIPDB, VirusTotal feeds      │
│ • Alert Engine: auto-creates alerts from anomaly scores     │
│ • Auth: JWT + RBAC (admin, soc_manager, analyst, viewer)    │
├──────────────────────────────────────────────────────────────┤
│ TIER 3: COMMAND CENTER (Next.js 16)                         │
│ • War Room dashboard (crown jewel — must be stunning)       │
│ • 10 modules total (see Section 5)                          │
│ • WebSocket client for real-time updates                    │
│ • Amharic/English bilingual (next-intl)                     │
│ • Hosted on Vercel, proxies API calls to VPS                │
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4. FILE STRUCTURE (FOLLOW EXACTLY)

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threatmatrix-ai/
├── docs/                          # Documentation
├── backend/
│   ├── app/
│   │   ├── main.py                # FastAPI entry point
│   │   ├── config.py              # Settings (pydantic BaseSettings)
│   │   ├── database.py            # SQLAlchemy engine + session
│   │   ├── dependencies.py        # Auth dependencies
│   │   ├── api/v1/                # Route modules (auth, flows, alerts, ml,
intel, llm, capture, reports, system)
│   │   ├── models/                # SQLAlchemy ORM models
│   │   ├── schemas/               # Pydantic request/response schemas
│   │   └── services/              # Business logic layer
│   ├── capture/                   # Scapy capture engine
│   │   ├── engine.py
│   │   ├── flow_aggregator.py
│   │   └── feature_extractor.py
│   ├── ml/
│   │   ├── datasets/              # Dataset loaders (nsl_kdd.py, cicids2017.py)
│   │   ├── models/                # Model implementations
│   │   ├── training/              # Training scripts
│   │   ├── inference/             # Real-time inference worker
│   │   └── saved_models/          # Serialized .pkl/.h5 files
│   ├── requirements.txt
│   ├── Dockerfile
│   └── alembic/
├── frontend/
│   ├── app/                       # Next.js App Router pages
│   │   ├── layout.tsx             # Root layout (dark theme, sidebar)
│   │   ├── war-room/page.tsx
│   │   ├── hunt/page.tsx
│   │   ├── intel/page.tsx
│   │   ├── network/page.tsx
│   │   ├── ai-analyst/page.tsx
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│   │   ├── alerts/page.tsx
│   │   ├── forensics/page.tsx
│   │   ├── ml-ops/page.tsx
│   │   ├── reports/page.tsx
│   │   └── admin/page.tsx
│   ├── components/                 # React components by module
│   ├── hooks/                      # Custom hooks (useWebSocket, useFlows, etc.)
│   ├── lib/                        # Utilities (api.ts, constants.ts)
│   ├── styles/globals.css          # Design system CSS variables
│   ├── messages/                   # i18n (en.json, am.json)
│   └── package.json
├── docker-compose.yml
├── docker-compose.dev.yml
├── nginx/nginx.conf
├── .env.example
└── Makefile
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5. MODULES (10 TOTAL — SCOPE LOCKED)

#	Module	Route	Priority	Description
1	**War Room**	`/war-room`	P0	Live threat map, metrics, alert feed, AI briefing, traffic timeline
2	**Threat Hunt**	`/hunt`	P0	Query builder, flow search, drill-down analysis
3	**Intel Hub**	`/intel`	P0	IOC browser, IP reputation, feed status, correlation
4	**Network Flow**	`/network`	P0	Traffic volume, top talkers, protocol breakdown, connection graph
5	**AI Analyst**	`/ai-analyst`	P0	Conversational LLM chat, streaming responses, quick actions
6	**Alert Console**	`/alerts`	P1	Alert table, lifecycle mgmt, severity filtering, bulk actions
7	**Forensics Lab**	`/forensics`	P1	PCAP upload, analysis, packet inspector, timeline
8	**ML Operations**	`/ml-ops`	P1	Model comparison, confusion matrix, ROC, feature importance
9	**Reports**	`/reports`	P1	PDF generation, daily briefing, incident reports
10	**Administration**	`/admin`	P2	User mgmt, config, LLM budget, feeds, audit log

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6. ML MODELS (3 MODELS — ENSEMBLE)

Model	Type	Library	Role

Model	Type	Library	Role
Isolation Forest	Unsupervised	scikit-learn	Zero-day anomaly detection
Random Forest	Supervised (multi-class)	scikit-learn	Known attack classification (Normal/DoS/Probe/R2L/U2R)
Autoencoder	Deep Learning (unsupervised)	TensorFlow/Keras	Complex pattern deviation

Ensemble Score: `0.30 × IF_score + 0.45 × RF_confidence + 0.25 × AE_error`
Datasets: NSL-KDD (primary), CICIDS2017 (secondary validation)

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7. LLM PROVIDERS (COST-OPTIMIZED)

Provider	Model	Use Case	Cost
DeepSeek	V3	Complex analysis, reasoning	$0.14/M tokens
Groq	Llama 3.3 70B	Real-time alerts (fastest)	$0.06/M tokens
GLM (Zhipu)	GLM-4-Flash	Bulk/translation (cheapest)	$0.01/M tokens

Total Budget: $100-200 | **Projected monthly usage:** ~$0.50

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8. DATABASE TABLES

`users`, `network_flows`, `alerts`, `threat_intel_iocs`, `ml_models`, `capture_sessions`, `pcap_uploads`, `llm_conversations`, `system_config`, `audit_log`

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9. UI DESIGN SYSTEM

- **Theme:** Dark only — `#0a0a0f` base, `#111118` panels, `#1a1a24` cards
- **Accent:** Cyber cyan `#00f0ff` (primary), `#ef4444` (critical), `#f97316` (high), `#22c55e` (safe)
- **Fonts:** `JetBrains Mono` (data/metrics), `Inter` (UI labels)
- **Style:** Glassmorphism panels, backdrop blur, scan-line animations, glow effects
- **Vibe:** Intelligence agency war room — NOT a SaaS dashboard

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10. CURRENT SPRINT (WEEK 1: Feb 24 — Mar 2)

Task	Owner	Status
Monorepo init, Docker Compose, CI	Lead	🔨 Today
PostgreSQL schema + Alembic migrations	Lead	📋 This week
FastAPI skeleton: auth, health, CORS	Lead	📋 This week

Task	Owner	Status
Next.js 16 init: layout, sidebar, theme	Full-Stack	📋 This week
Redis setup + pub/sub test	Lead	📋 This week
Design system CSS	Full-Stack	📋 This week

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11. REFERENCE DOCUMENTS

All detailed specifications live in `/docs/`:

- `MASTER_DOC_PART1_STRATEGY.md` — Business case, market, risks
- `MASTER_DOC_PART2_ARCHITECTURE.md` — DB schema, API spec, security
- `MASTER_DOC_PART3_MODULES.md` — Module specs, UI/UX design system
- `MASTER_DOC_PART4_ML_LLM.md` — ML pipeline, LLM gateway, prompts
- `MASTER_DOC_PART5_TIMELINE.md` — Sprint plan, deployment, testing

When you need detail on any section, ask for the specific Part number.

END OF GLOBAL CONTEXT — Any AI model receiving this document has full authority to assist with ThreatMatrix AI implementation within the constraints defined above.